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9.1 醇的分类、结构、命名和物理性质

醇是羟基连接在饱和 sp^3 碳原子上的化合物

醇

酚

烯醇

- 醇可看成是水分子中的一个氢原子被烃基取代所形成的有机衍生物
- 醇的官能团是羟基(hydroxyl group) $-OH$

9.1.1 醇的分类

3种分类方法:

1) 根据官能团所连烃基类型

RCH_2-OH	R_2CH-OH	R_3C-OH
伯醇	仲醇	叔醇
一级醇(1°)	二级醇(2°)	三级醇(3°)

9.1.1 醇的分类

2) 根据烃基结构

饱和醇

不饱和醇

芳香醇

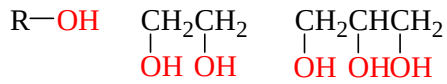


3) 根据羟基数目

一元醇

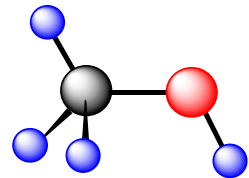
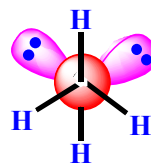
二元醇

多元醇



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9.1.2. 醇的结构

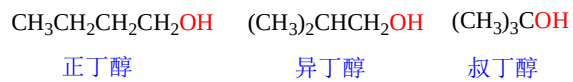
C、O都是 sp^3 杂化

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9.1.3 醇的命名

1) 普通命名

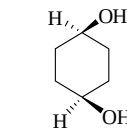
中文：将相应烷烃名称中的烷→醇



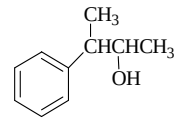
7

9.1.3 醇的命名

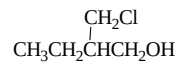
2) 系统命名



顺-1,4-环己二醇



3-苯基-2-丁醇



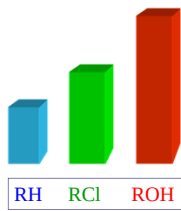
2-(氯甲基)-1-丁醇

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9.1.4 物理性质

• 醇的沸点

比同碳原子数的烷烃、卤代烃高，也比分子量相近的烷烃高

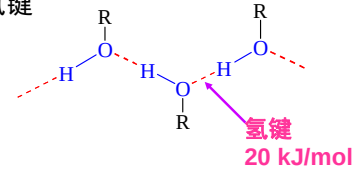


化合物	分子量	沸点(°C)
CH ₃ CH ₂ OH	46	78.5
CH ₃ CH ₃	30	-98
CH ₃ CH ₂ CH ₃	44	-42
CH ₃ CH ₂ Cl	64.5	12

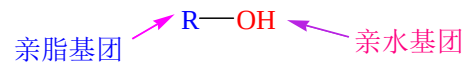
9

9.1.4 物理性质

• 醇的分子间氢键

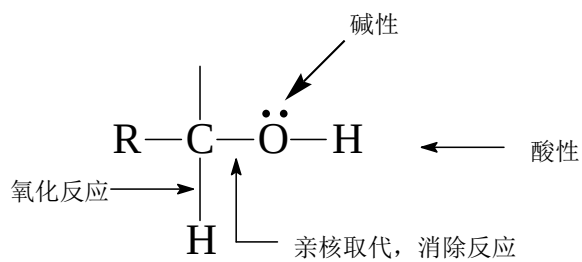


• 溶解度：低级醇溶于水，随分子量增加溶解度降低

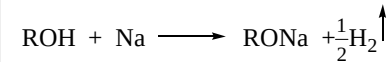
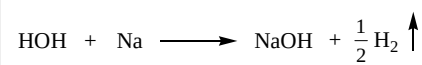


10

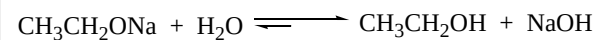
醇的化学性质



醇的酸性



反应速度 $\text{CH}_3\text{OH} > \text{CH}_3\text{CH}_2\text{OH} > \text{CH}_3\text{CH}_2\text{CH}_2\text{OH} > (\text{CH}_3)_2\text{CHOH} > (\text{CH}_3)_3\text{COH}$



醇是一种弱酸

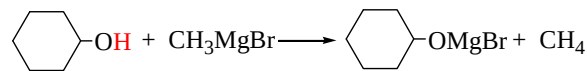
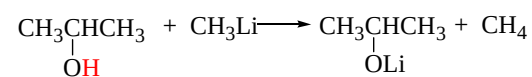
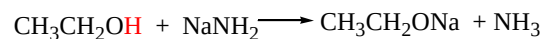
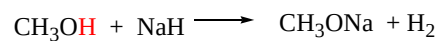
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酸性大小

化合物	pK _a	
(CH ₃) ₃ COH	18.00	弱酸
CH ₃ CH ₂ OH	16.00	
HOH	15.74	
CH ₃ OH	15.54	
CF ₃ CH ₂ OH	12.43	
(CF ₃) ₃ COH	5.4	
HCl	-7.00	强酸

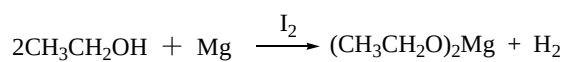
13

Example

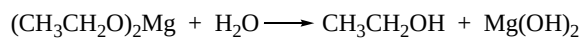


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醇与Mg反应



乙醇镁

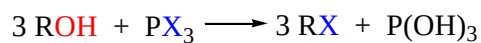
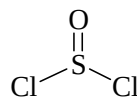
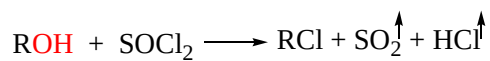


用于制备高度无水乙醇

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取代反应

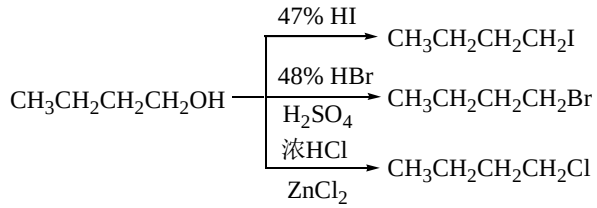
1. 与三卤化磷和氯化亚砷反应

或PX₅ 或X₂, P X = Br, I氯化亚砷
亚硫酸氯

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取代反应

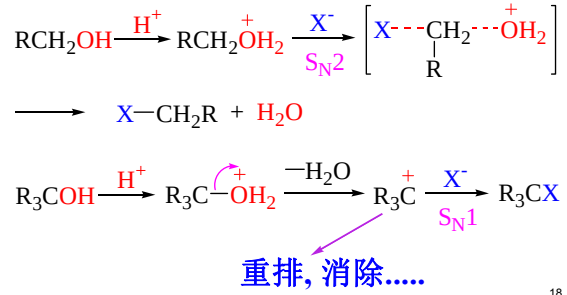
2. 与氢卤酸反应



反应活性: $\text{HI} > \text{HBr} > \text{HCl}$
 $3^\circ \text{ 醇} > 2^\circ \text{ 醇} > 1^\circ \text{ 醇}$

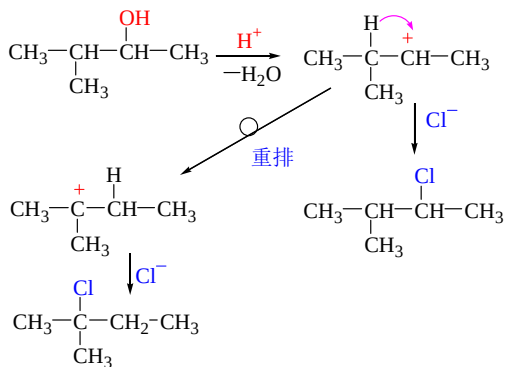
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反应机理



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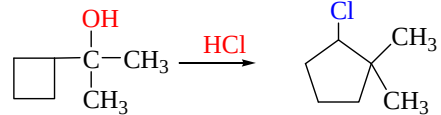
反应机理



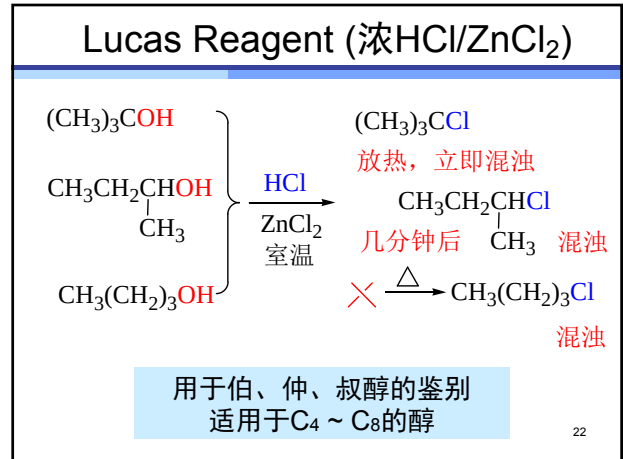
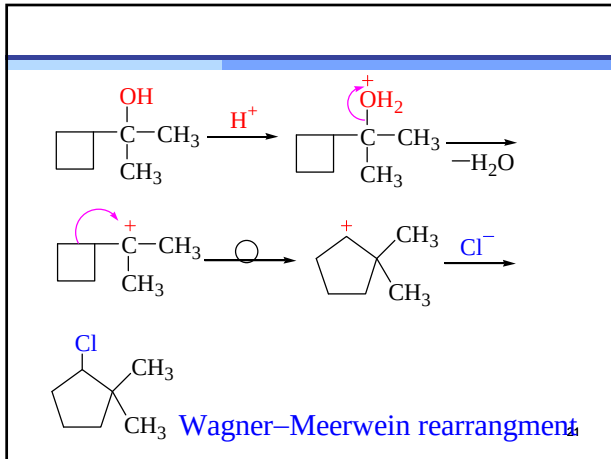
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测试

请同学们用机理解释如下反应



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用于伯、仲、叔醇的鉴别
适用于C₄ ~ C₈的醇

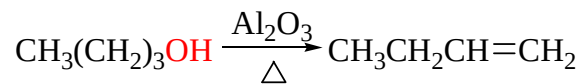
- 关键在生成比水重的卤代烃
- 氯代丙烷, 密度0.89 g/mL
- 氯代异丙烷, 密度0.868 g/mL
- 高级醇难溶于水

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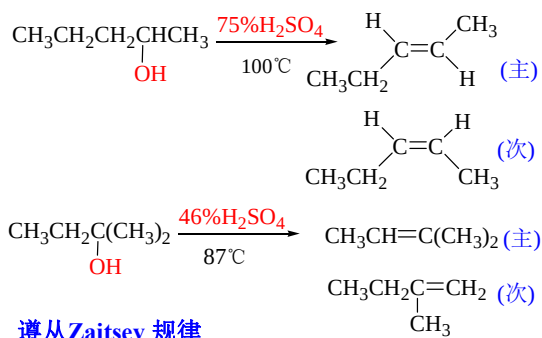
脱水反应

常用催化剂: H₂SO₄, H₃PO₄, Al₂O₃

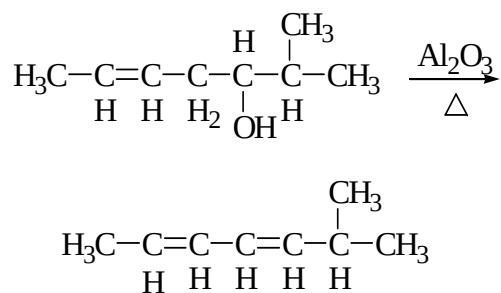
1. 分子内脱水



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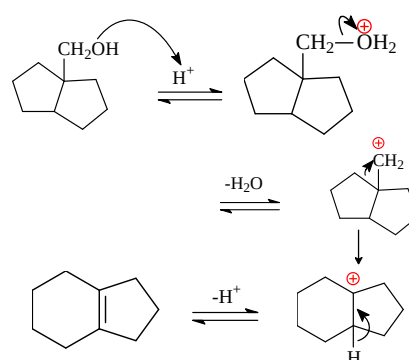
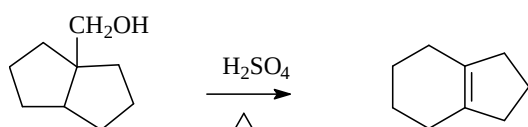


共轭规律优先于Zaitsev规律

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Example

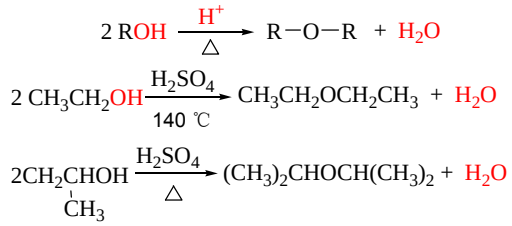
请同学们用机理
解释如下反应



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脱水反应

2. 分子间脱水

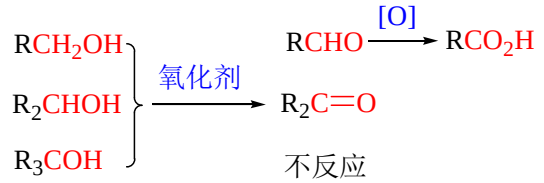


主要副反应：分子内脱水。
不适用于三级醇

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氧化与脱氢

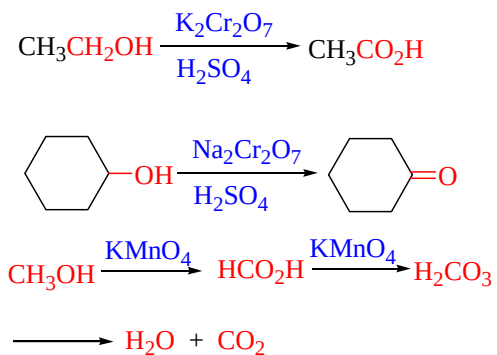
• 氧化剂氧化



氧化剂：KMnO₄, H₂CrO₄, K₂Cr₂O₇/H₂SO₄
Na₂Cr₂O₇/H₂SO₄, CrO₃/H₂SO₄

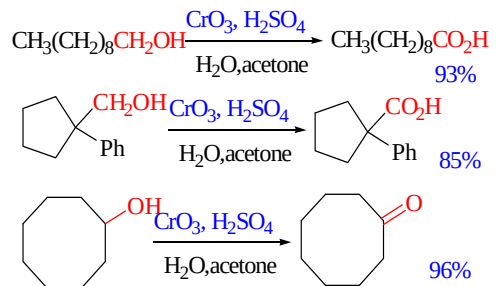
30

Example



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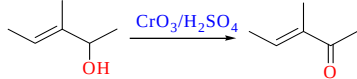
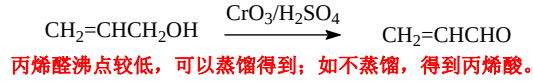
Jones' reagent — CrO₃ / H₂SO₄



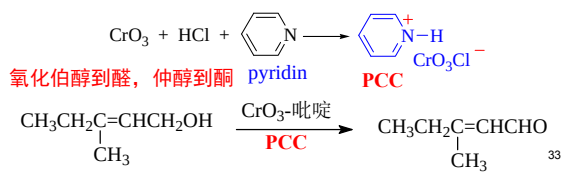
• 氧化伯醇到酸，仲醇到酮，烯丙醇、苄醇到醛

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Johns试剂: $\text{CrO}_3/\text{H}_2\text{SO}_4$



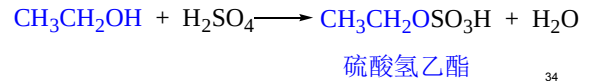
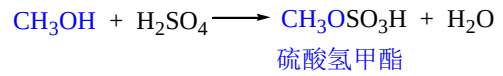
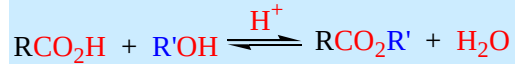
PCC (Pyridinium Chlorochromate)



33

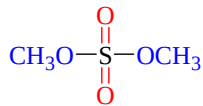
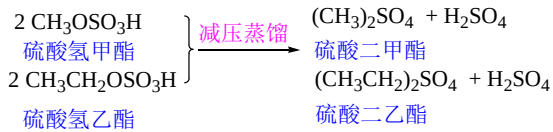
成酯反应

醇与含氧酸(有机酸、无机酸)反应成酯



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用途

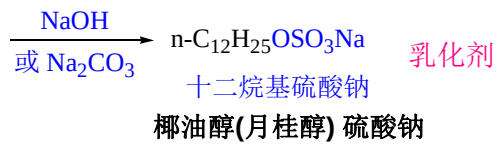
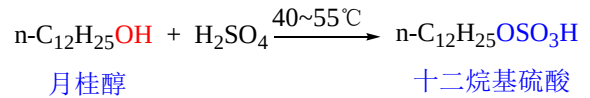


烷基化试剂

剧毒!

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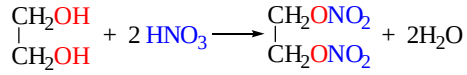
用途



阴离子型表面活性剂、乳化剂及发泡剂

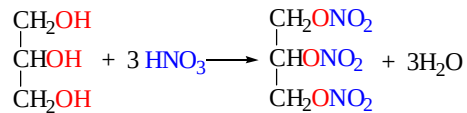
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用途



乙二醇

乙二醇二硝酸酯



甘油

甘油三硝酸酯

有扩张冠脉血管的作用，
可以缓解狭窄、痉挛的冠
脉血管，改善心肌循环。

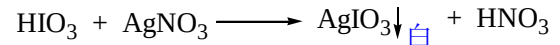
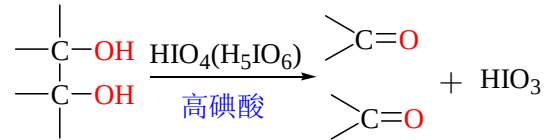
硝化甘油

烈性炸药

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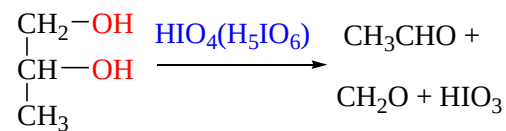
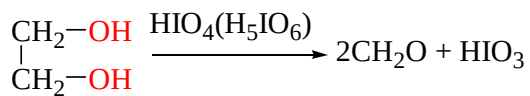
邻二醇的反应

1. 被高碘酸氧化



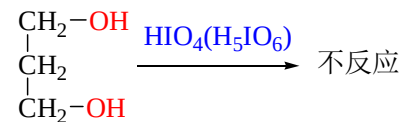
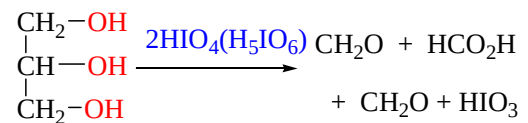
38

Example



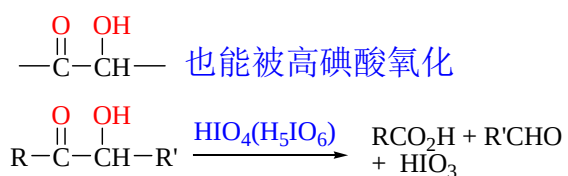
39

Example



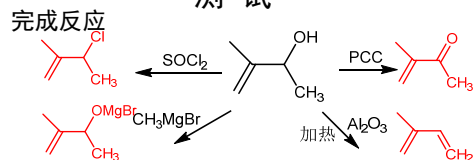
40

Example

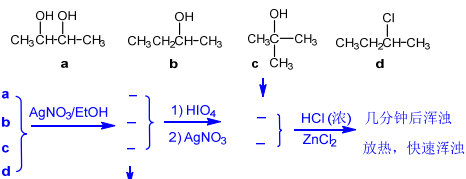


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测试



用化学方法鉴别如下化合物

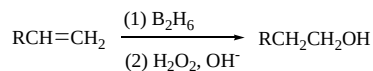
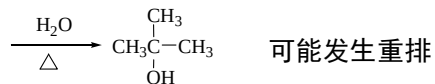
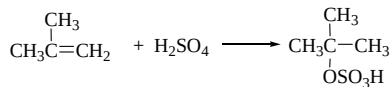
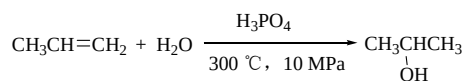


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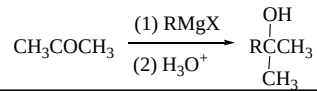
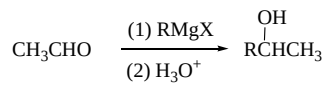
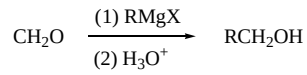
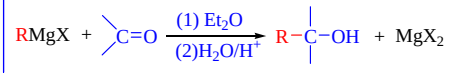
醇的制备和来源

- 由烯烃制备——水合，硼氢化-氧化
- 由羰基化合物制备——Grignard 反应，与炔化物反应，还原
- 由卤代烃水解

由烯烃制备

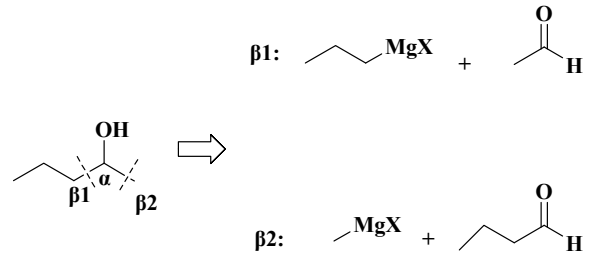


由羰基化合物制备



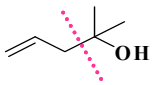
Example

逆向合成分析



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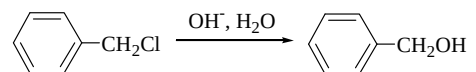
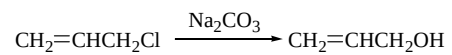
Example



由卤代烃水解

- 副反应多
- 一般情况下醇比卤代烃易得

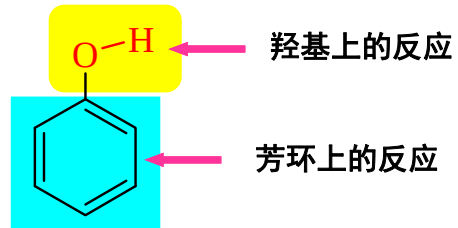
由卤代烃水解主要应用到如下类型的化合物中：



9.4 酚的结构、命名、来源 和物理性质(自学)

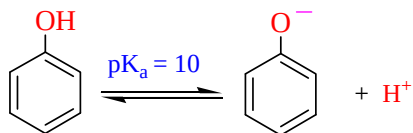
- 分类：根据分子中羟基的数目分为一元酚，二元酚和多元酚
- 命名：一般以苯酚为母体命名
- 物理性质：具有特殊气味，能形成分子间氢键，沸点较高，水中有一定溶解度

酚的化学性质



羟基上的反应

1. 酚的酸性

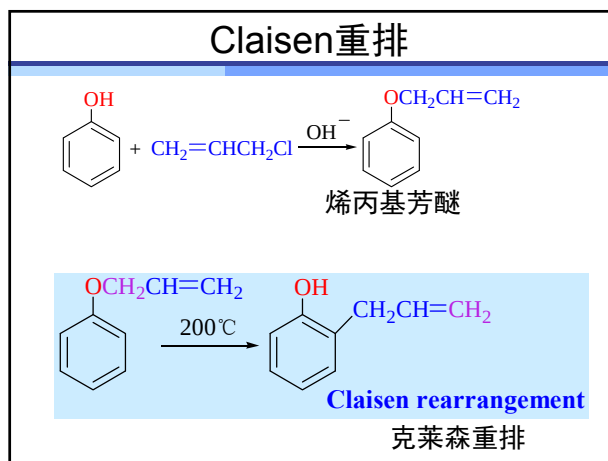
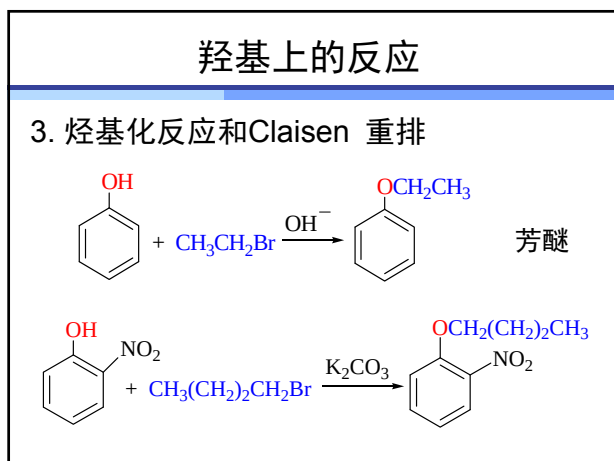
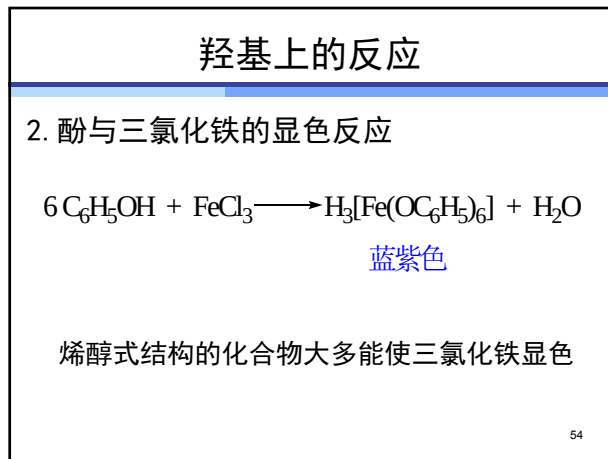
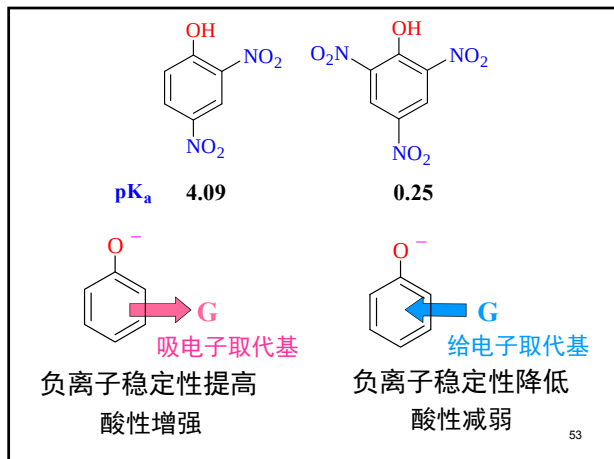


取代酚的酸性

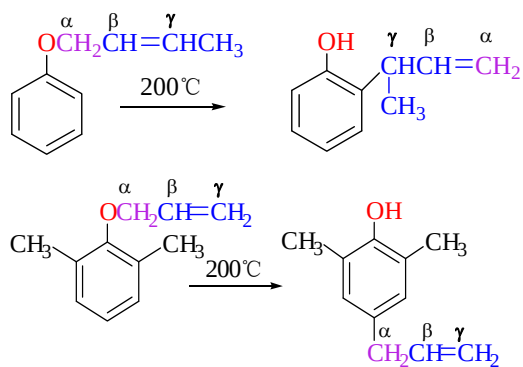
G	NH ₂	CH ₃ O	CH ₃	H	Cl	Br	I	NO ₂	
pK _a	10.46	10.21	10.17	10.0	9.38	9.35	9.30	7.15	
	NO ₂	<i>m</i> -	<i>o</i> -	<i>p</i> -					
pK _a	8.28	7.17	7.15						

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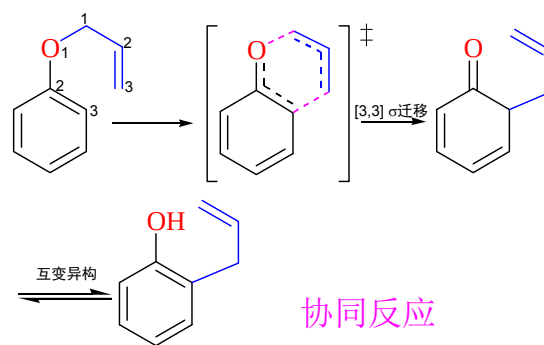
52



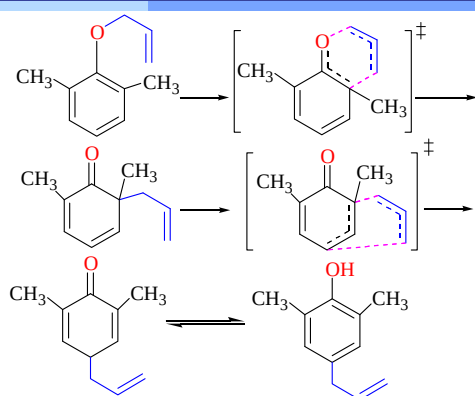
Example



反应机理



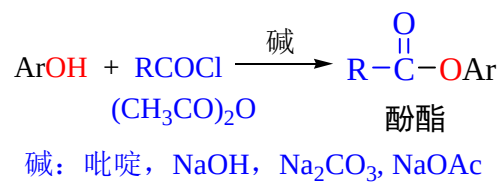
反应机理



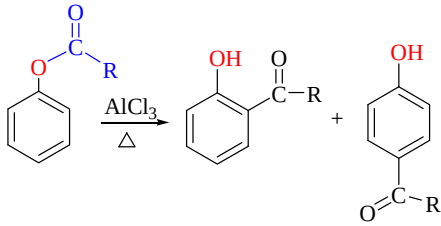
羟基上的反应

4. 酰基化反应及Fries重排

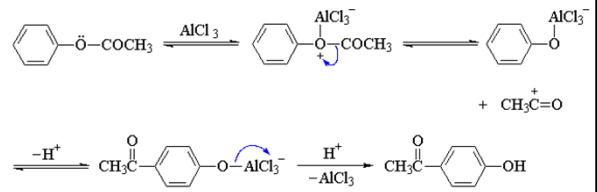
一般用酰氯或酸酐反应



Fries重排

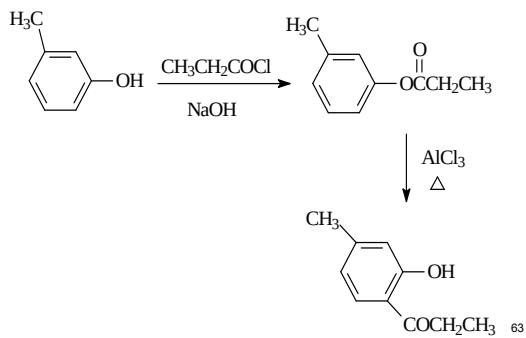


高温下产物以邻位为主
较低温度下产物以对位为主



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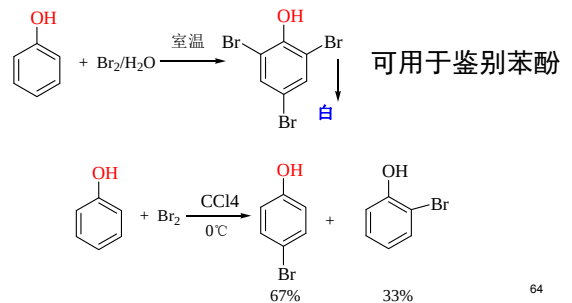
Example



63

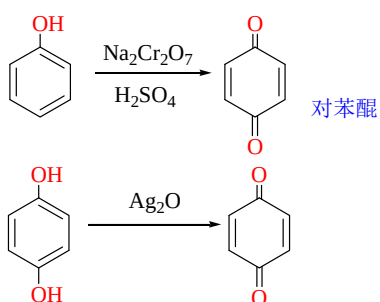
芳环上的取代反应(自学)

卤化反应



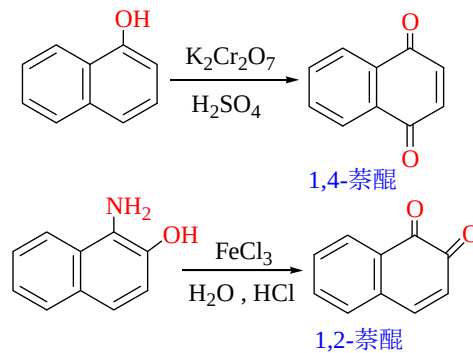
64

酚的氧化反应



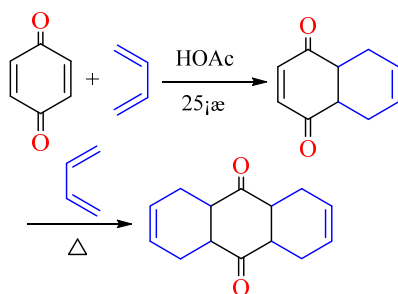
65

Example



66

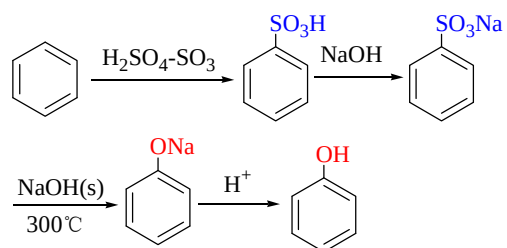
对苯醌可做为亲双烯体
发生Diels—Alder反应



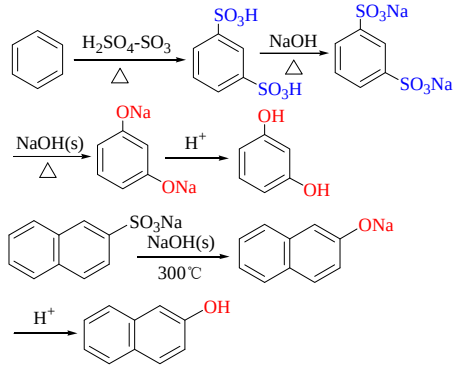
67

酚的制备

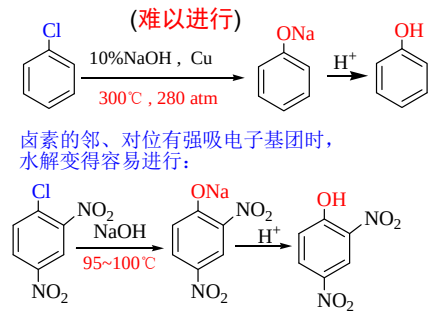
1. 磺酸盐碱熔法



Example

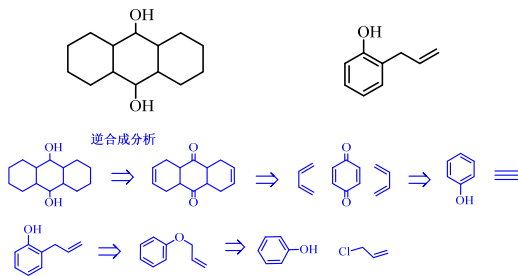


2. 氯苯水解法



测试

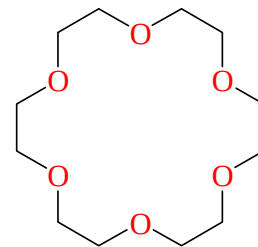
以苯酚位原料，合成下列物质



71

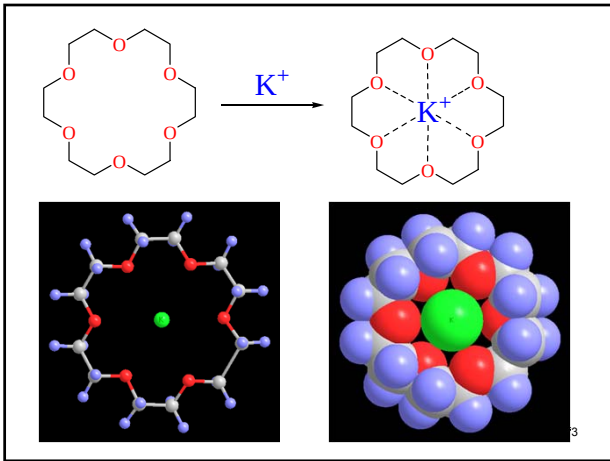
大环多醚

$(\text{OCH}_2\text{CH}_2)_n$ 冠醚(Crown ether)



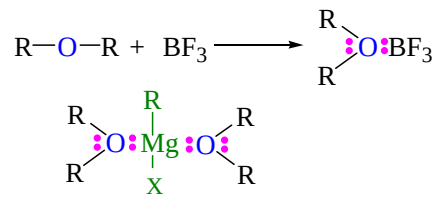
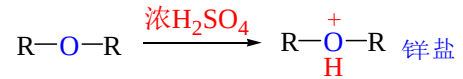
18-冠-6
18-crown-6

72



醚的化学反应

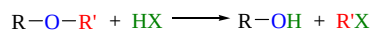
1. 形成铯盐



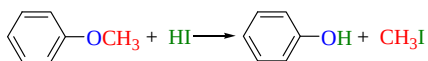
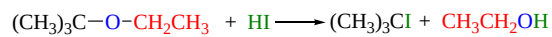
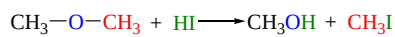
74

醚的化学反应

2. 醚键的断裂



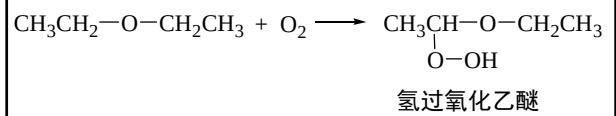
反应活性 $\text{HI} > \text{HBr} > \text{HCl}$



75

醚的化学反应

3. 过氧化物peroxide的生成

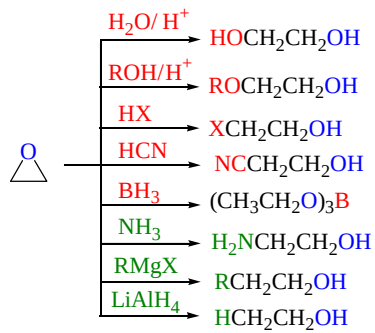


- 检验：淀粉—碘化钾试纸
- 除去过氧化物的方法： FeSO_4 洗涤
- 储存时加Fe可防止形成过氧化物

76

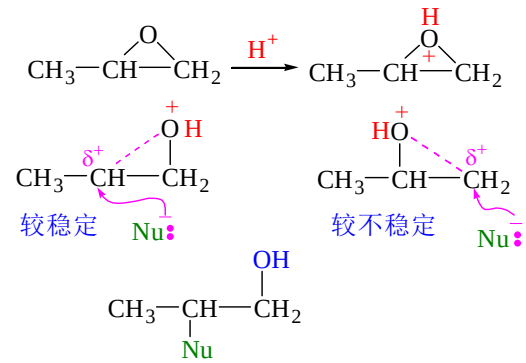
醚的化学反应

4. 环氧化合物的反应



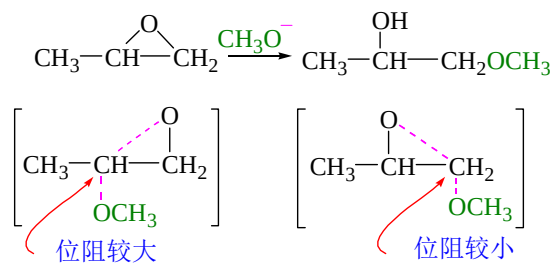
77

酸催化下的开环反应



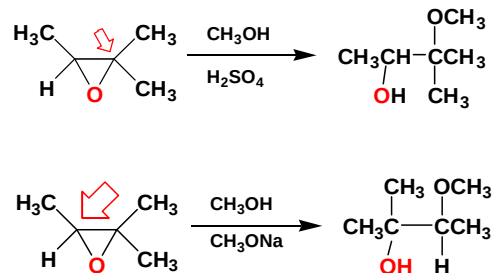
78

碱催化下的开环反应



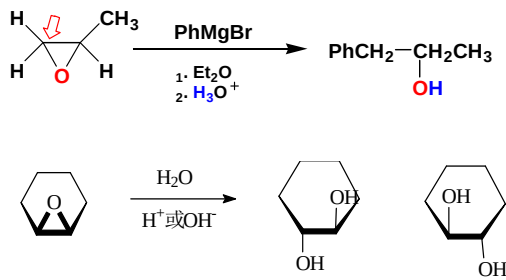
79

Example



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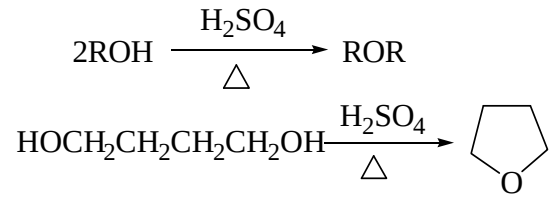
Example



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醚的制备

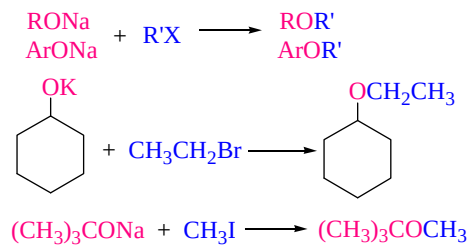
1. 醇的分子间脱水



82

醚的制备

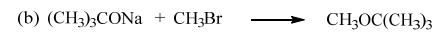
2. Williamson 合成法



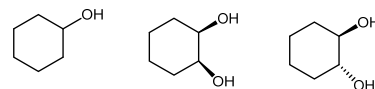
83

测试

1、下列两个制备甲基叔丁基醚的反应，哪个产率高？

**b) 产率高；a) 消除为主**

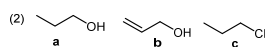
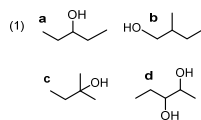
2、用环己烯制备如下化合物



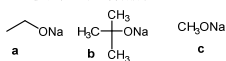
84

作业(一)

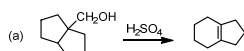
1. 用简单的化学方法鉴别下列各组化合物



2. 按碱性由强到弱排序



3. 写出下列反应机理



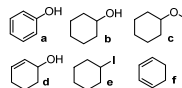
4. 以C2及以下有机物合成下列物质



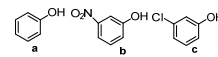
85

作业(二)

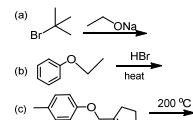
5. 用简单的化学方法鉴别下列6个化合物



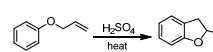
6. 排序题 按酸性由强到弱排序



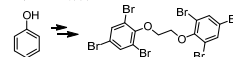
7. 完成下列反应



8. 写出下列反应机理



9. 完成下列转换



10. 以苯、C3及以下有机物合成下列物质



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